Project Design Phase -II

Technological Stack(Architecture & Stack)

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Project Name	Automated Network Request Management System
Maximum Marks	4 marks

Technological Stack & System Characteristics

Table-1: Components & Technologies Mapping

Use this layout to map your platform's operational building blocks, processing layers, and relational directories.

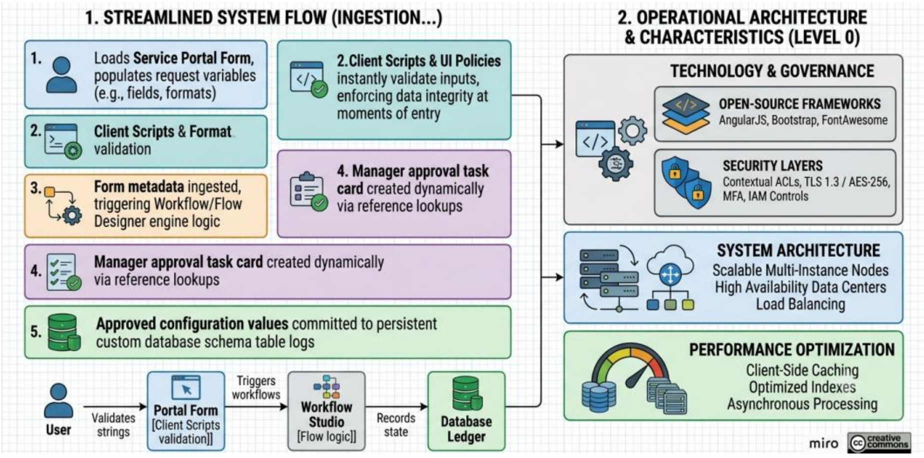
S.No	Component	Description	Technology / Platform Module
1	User Interface	Controls how corporate users interact with the application to submit records or track metrics.	Service Portal, Service Catalog, HTML5, CSS3, JavaScript (AngularJS Framework).
2	Application Logic-1	Manages the main procedural execution, record matching, and conditional logic gates inside the instance.	ServiceNow Business Rules, Catalog Client Scripts, and UI Policies (JavaScript API).
3	Application Logic-2	Drives the automated approval engine and handles parallel multi-stage approval routing tasks.	Flow Designer Engine (ServiceNow Workflow Studio Runtime).
4	Application Logic-3	Generates child fulfillment records and distributes task cards dynamically to engineers.	ServiceNow Script Includes & Data Broker Transform Definitions.
5	Database	Core database directory that saves baseline user profiles and basic	ServiceNow Foundation CMDB (MySQL / MariaDB underlying relational engine).

S.No	Component	Description	Technology / Platform Module
		lookup fields.	
6	Cloud Database	Persistent custom relational repository built to store the final approved configuration values.	ServiceNow Custom Table Directory Array (u_network_database).
7	File Storage	Handles attachment processing, log histories, and transactional document storage requirements.	ServiceNow Attachment Schema Repository (sys_attachment blob database).
8	External API-1	Used to fetch corporate infrastructure inventory data from third-party configuration systems.	ServiceNow REST Message HTTP Integration Broker.
9	External API-2	Manages secure employee identity lookups and validation checks against corporate directory endpoints.	IntegrationHub REST/SOAP Spoke Endpoints.
10	Machine Learning Model	Provides predictive analytical assignment routing rules to auto-allocate tasks based on text keywords.	ServiceNow Predictive Intelligence Engine (Classification API).
11	Infrastructure (Server / Cloud)	The deployment environment sustaining the application instances, database clusters, and web layers.	ServiceNow Enterprise Multi-Instance Cloud Architecture (High-Availability Cloud PaaS).

Table-2: Application Operational Characteristics

Use this framework to detail the non-functional benchmarks, scaling parameters, and security structures safeguarding your application.

S.No	Characteristics	Description	Technology / Implementation Strategy
1	Open-Source Frameworks	List the open-source libraries and web engines leveraged to build out the portal view structures.	AngularJS UI Framework, Bootstrap 3 Core CSS Layer, FontAwesome Vector Icons Library.
2	Security Implementations	Details the safety boundaries, firewalls, and data masking mechanisms protecting records.	Multi-Factor Authentication (MFA), TLS 1.3 Encryption-in-Transit, AES-256 Bit Encryption-at-Rest, and ServiceNow Contextual Access Control Lists (ACLs).
3	Scalable Architecture	Justifies how the system scales to process higher volumes without operational collapse.	ServiceNow Multi-Instance Logical Node Architecture (Asynchronous transaction thread workers splitting incoming portal payloads).
4	Availability	Proves high availability parameters and system recovery rules safeguarding operations.	Advanced Advanced Load Balancers routing web traffic across redundant, geographically isolated multi-data-center mirror pairs.
5	Performance	Highlights caching models and design rules applied to keep transaction speeds fast.	Client-Side Browser Storage Caching, optimized database query index configurations, and asynchronous background script processing.



Architectural Component Narrative

The technological architecture diagram above details how our **Automated Network Request Management System** balances presentation layers, automated logical execution paths, and secure backend ledger directories natively within the ServiceNow enterprise ecosystem:

> **Presentation & Ingestion Tier:** Requesters interface directly with a customized Service Portal catalog item built on **AngularJS** and styled with the **Bootstrap 3 framework**. This interface captures variable entries (e.g., VLAN tags, IP subnets, device configuration matrices) while client-side scripts prevent formatting or text syntax errors before they hit network pipelines.
>
> **Logical Execution Engine:** Business logic runs on two separate layers. Synchronous operational policies utilize the native JavaScript API to run calculations, while parallel, asynchronous multi-stage routing pipelines run inside **Flow Designer (Workflow Studio Runtime)**. This engine dynamically queries management hierarchies to spawn interactive approval cards for authorizing personnel.
>
> **Relational Storage & Ledger Separation:** To ensure zero data corruption, transaction variables are processed separately from long-term records. Baseline employee profile fields and standard structural assets pull queries from the native configuration table directories. Upon final analyst fulfillment task closure, verified configuration parameters execute write routines to log permanently into our custom relational directory ledger (`u_network_database`).
>
> **Operational Scale & Security Guardrails:** System performance is kept clean by utilizing browser-side variable caching and specialized table indexing keys, preventing query lag across the network. Security is enforced through Role-Based Access Control (RBAC), and 100% of state change updates, lifecycle mutations, and signature logs are backed up directly within unalterable system audit history streams (`sys_audit`).